FRE-GY 9743 Assignment 5 (Math Part)

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Q1. Solving $V(X_I) = X_M$

Since $\dim(X_I) = \dim(X_M)$, this is a square system of equations. Define the residual

$$R(X_I) = V(X_I) - X_M = 0.$$

1) Initial guess. Start from a sensible $X_I^{(0)}$ (e.g. yesterday's calibration or a flat/naive guess).

2) Newton–Raphson. With the Jacobian $J(X_I) = \dfrac{\partial V}{\partial X_I}$ (square, by AD or finite differences), iterate

$$X_I^{(n+1)} = X_I^{(n)} - J\big(X_I^{(n)}\big)^{-1} R\big(X_I^{(n)}\big)$$

until $\|R\| < \varepsilon$. In practice one solves the linear system $J\,\Delta = -R$ rather than inverting J.

3) Robustness. If Newton is unstable (bad guess, near-singular J), use a damped step / line search, or minimize $\frac{1}{2}\|R\|^2$ with Levenberg–Marquardt. A unique exact root requires J to be invertible near the solution.

Q2. Sensitivity dX_I/dX_M at the Solution

At the solution X_I^* the identity $V\big(X_I^*(X_M)\big) = X_M$ holds for every X_M. Differentiating both sides with respect to X_M (implicit function theorem):

$$\left.\frac{\partial V}{\partial X_I}\right|_{X_I^*} \frac{dX_I}{dX_M} = I \quad \Longrightarrow \quad \frac{dX_I}{dX_M} = J\big(X_I^*\big)^{-1}.$$

No re-calibration (bump-and-resolve) is needed: the Jacobian from the last Newton step already gives the answer. More generally, for $F(X_I, X_M) = 0$,

$$\frac{dX_I}{dX_M} = -\left(\frac{\partial F}{\partial X_I}\right)^{-1} \frac{\partial F}{\partial X_M}.$$

This only requires J to be invertible at X_I^*.

Q3. Calibrating a 3-Point Vol Interpolator

1) Mapping to the recipe. $X_I = \theta \in \mathbb{R}^3$, $X_M = (P_1, P_2, P_3)$, and

$$V_i(\theta) = \mathrm{Black}\big(F, K_i, T, \sigma(K_i;\theta)\big), \qquad i = 1, 2, 3,$$

the un-discounted Black price (put for K_1, call for K_3, either at K_{atm}). Three equations, three unknowns.

2) Finding θ. Solve $V(\theta) = P$ with Newton as in Q1. The Jacobian factorizes through the chain rule,

$$J_{ij} = \frac{\partial V_i}{\partial \theta_j} = \mathcal{V}_i \,\frac{\partial \sigma(K_i;\theta)}{\partial \theta_j}, \qquad \mathcal{V}_i = \left.\frac{\partial\, \mathrm{Black}}{\partial \sigma}\right|_{K_i}.$$

Since vega $\mathcal{V}_i > 0$, each price is monotone in its vol, so the solution is unique as long as $\partial\sigma(K_i;\theta)/\partial\theta$ is invertible. A better-conditioned alternative is to first invert each price to a market implied vol σ_i^{mkt} and solve $\sigma(K_i;\theta) = \sigma_i^{mkt}$ (removes the vega scaling).

3) How $\sigma(K)$ moves. For fixed market prices, $\partial\sigma(K;\theta)/\partial\theta$ is explicit from the interpolator. The sensitivity to the market quotes follows from Q2:

$$\frac{d\sigma(K)}{dP} = \frac{\partial\sigma(K;\theta)}{\partial\theta}\,\frac{d\theta}{dP} = \frac{\partial\sigma(K;\theta)}{\partial\theta}\,J(\theta^*)^{-1}.$$

This gives the vol at any strike (and hence the risk of any option priced off the surface) in terms of the three quoted instruments.

4) Intuitive θ. Raw parameters (e.g. polynomial or SABR coefficients) mean little to a trader. Better to parametrize the smile by what traders quote:

- **Level:** ATM vol σ_{atm}.
- **Skew:** risk reversal $\sigma(K_3) - \sigma(K_1)$.
- **Convexity:** butterfly $\frac{1}{2}\big(\sigma(K_1) + \sigma(K_3)\big) - \sigma_{atm}$.

Equivalently, take $\theta = (\sigma_1, \sigma_{atm}, \sigma_3)$, the pillar vols themselves. Then $\partial\sigma(K)/\partial\theta_j$ is just the interpolation weight on pillar j, and risk reads directly as bucketed vega against the three traded options.